

Name: _____ Class: _____

WHITEBRIDGE HIGH SCHOOL



2006

Higher School Certificate

Trial HSC Examination

MATHEMATICS EXTENSION 2

Time Allowed: 3 hours

(Reading time: 5 minutes)

Directions to Candidates

- All questions of equal value.
- Commence each question on a new page.
- Marks may be deducted for careless or badly arranged work.

Question 1 (15 marks) Commence each question on a SEPARATE page

a. Find $\int \frac{dx}{x \log_e x}$ **2**

b. Find $\int \frac{dx}{\sqrt{3+2x-x^2}}$ **2**

c. Find $\int \frac{dx}{(x+1)(x^2+4)}$ **4**

d. Using the substitution $t = \tan \frac{x}{2}$, calculate $\int \frac{15}{17+8\cos x} dx$, leaving your answer in terms of t . **3**

e. i. Differentiate $\frac{x}{\sqrt{x-3}}$. **1**

ii. Hence evaluate $\int_4^7 \frac{2x-9}{2(x-3)\sqrt{x-3}} dx$ **3**

Question 2 (15 marks) Commence each question on a SEPARATE page

- a. i. Express $-1 + i\sqrt{3}$ in modulus argument form. **2**
- ii. Hence evaluate $(-1 + i\sqrt{3})^{-6}$ **2**
- b. If z is a non-zero complex number such that $z + \frac{1}{z}$ is real, prove that **3**
 $\operatorname{Im}(z) = 0$ or $|z| = 1$.
- c. Sketch the region where the inequalities $-\frac{\pi}{2} \leq \arg(z - 1 - 2i) \leq \frac{\pi}{4}$, and $|z| \leq \sqrt{5}$ **3**
both hold.
- d. Let z be a complex number for which $|z| = 1$ and $\arg z = \theta$, where $0 < \theta < \frac{\pi}{2}$.
- i. Show that $|1 - z| = \sqrt{2 - 2\cos\theta}$ and $|1 + z| = \sqrt{2 + 2\cos\theta}$ **3**
- ii. Hence find the value of $\left| \frac{2}{1 - z^2} \right|$ in terms of θ . **2**

Question 3 (15 marks) Commence each question on a SEPARATE page

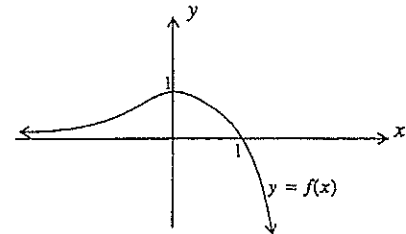
- a. Show that $\int_0^{\frac{\pi}{4}} x \sin x \, dx = \frac{\sqrt{2}}{8} (4 - \pi)$ **3**
- b. The shape of a particular cake can be represented by rotating the region **4**
between the curve $y = \sin x$ and the x -axis, from $x = 0$ and $x = \frac{\pi}{4}$, about the
line $x = \frac{\pi}{4}$. Using the method of cylindrical shells, find the volume of the cake.
- c. The hyperbola H has equation $9x^2 - 4y^2 = 36$.
- i. Find the co-ordinates of the foci, S and S' . **2**
- ii. Find the equations of the directrices. **1**
- iii. Find the equations of the asymptotes. **1**
- iv. Sketch the curve H indicating the information obtained in i. to iii. **1**
- v. The point $P(x_0, y_0)$ lies on H . Prove that the equation of the tangent at P **3**
is $9x_0x - 4y_0y = 36$.

Question 4 (15 marks) Commence each question on a SEPARATE page

- a. The graph of
- $y = f(x)$
- is sketched below.

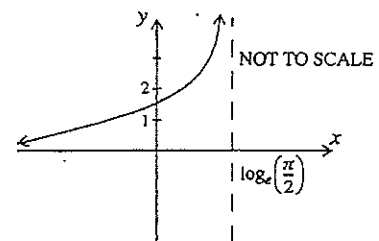
There is a stationary point at $(0, 1)$

Use this graph to sketch the following without calculus, showing essential features.



- | | | |
|------|---------------------------------|----------|
| i. | $y = f\left(\frac{x}{2}\right)$ | 1 |
| ii. | $y = x + f(x)$ | 2 |
| iii. | $y = \frac{1}{f(x)}$ | 2 |
| iv. | $y = f\left(\frac{1}{x}\right)$ | 2 |

- b. The diagram shows part of the curve $y = \tan(e^x)$, where $x < \log_e\left(\frac{\pi}{2}\right)$. The part to the right of $\log_e\left(\frac{\pi}{2}\right)$ has not yet been drawn.



- | | | |
|------|--|----------|
| i. | By considering values of x greater than $\log_e\left(\frac{\pi}{2}\right)$, find the smallest positive solution to the equation $\tan(e^x) = 0$. | 1 |
| ii. | Copy the diagram and hence sketch the curve $y = \tan(e^x)$ for $x < \log_e\left(\frac{3\pi}{2}\right)$. | 1 |
| iii. | How many solutions are there to the equation $\tan(e^x) = 0$ in the domain $1 < x < 3$? | 2 |
| iv. | Find the equation of the inverse function of the $y = \tan(e^x)$ for the case when | |
| α. | $x < \log_e\left(\frac{\pi}{2}\right)$. | 2 |
| β. | $\log_e\left(\frac{\pi}{2}\right) < x < \log_e\left(\frac{3\pi}{2}\right)$ | 2 |

Question 5 (15 marks) Commence each question on a SEPARATE page

- a. i. Prove that $\tan^{-1} n - \tan^{-1}(n - 1) = \tan^{-1} \frac{1}{n^2 - n + 1}$, where n is a positive integer. **2**
- ii. Hence evaluate $\tan^{-1} 1 + \tan^{-1} \frac{1}{3} + \dots + \tan^{-1} \frac{1}{n^2 - n + 1}$. **2**
- iii. Hence find the limit $\sum_{n=1}^{\infty} \tan^{-1} \frac{1}{n^2 - n + 1}$. **1**
- b. A food package of mass m kg has a parachute device attached. It is released from rest from the top of a cliff 100 metres high. During its fall, the only forces acting are gravity, and owing to the parachute, a resistive force of magnitude $\frac{1}{10}mv^2$, where v metres per second is the speed of the package
- After $\frac{1}{2} \ln 99$ seconds, the parachute disintegrates, and then the only force acting on the particle is due to gravity.
- The acceleration due to gravity is taken as 10 m s^{-1} . At time t seconds after being dropped, the package has fallen a distance of x metres from the plane, and its speed is $v \text{ m s}^{-1}$.
- i. Show that while the parachute is operating, $\ddot{x} = 10 - \frac{v^2}{10}$. Hence show **5**
that $v = 10 \left(\frac{e^{2t} - 1}{e^{2t} + 1} \right)$ and $x = 5 \ln \left(\frac{100}{100 - v^2} \right)$
- ii. Find the exact speed of the package and the exact vertical distance fallen just before the parachute disintegrates. **2**
- iii. Find the speed of the package just before it reaches the ground. **3**
Give your answer correct to two significant figures.

Question 6 (15 marks) Commence each question on a SEPARATE page

- a. The polynomial $P(z)$ has equation $P(z) = z^4 - 2z^3 - z^2 + 2z + 10$ **3**
 Given that $z - 2 + i$ is a factor of $P(z)$, express $P(z)$ as a product of two quadratic factors with real coefficients.

- b. A particle moves in a straight line. It is placed at the origin on the x -axis and is then released from rest. When at position x , its acceleration is given by

$$\ddot{x} = -9x + \frac{5}{(2-x)^2}.$$

- i. Show that $v^2 = \frac{x(3x-5)(3x-1)}{2-x}$. **2**
- ii. Prove that the particle moves between two points on the x -axis, and find these points. **4**

- c. An athlete is throwing a javelin. The horizontal and vertical components of the speed of the javelin after t seconds are:

$$\dot{x} = V + 3V\cos\theta \text{ and } \dot{y} = 3V\sin\theta - gt$$

where V is a positive constant, θ is an acute angle, and x and y are the horizontal and vertical displacements from the point of projection.

(Assume when $t = 0$, $x = 0$ and $y = 0$)

Show that:

- i. $x = Vt + 3Vt\cos\theta$ and $y = 3Vt\sin\theta - \frac{1}{2}gt^2$. **2**
- ii. the range of the javelin, R metres, is given by $R = \frac{6V^2 \sin\theta}{g} (3\cos\theta + 1)$. **2**
- iii. the angle θ which will yield maximum range is $\theta = \cos^{-1}\left(\frac{\sqrt{73}-1}{12}\right)$. **3**

Question 7 (15 marks) Commence each question on a SEPARATE page

- a. Given that the quartic polynomial $p(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$ has a zero of multiplicity three, factorise the polynomial completely and find all its zeroes. **3**

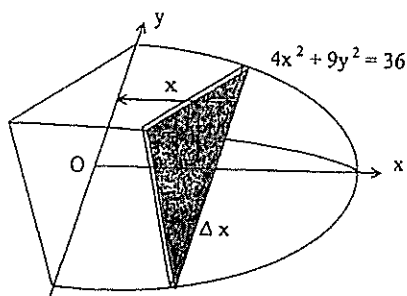
- b. Let $Q(x) = x^3 + px + q$, where p and q are real and non-zero. Two of the zeroes of $Q(x)$ are $a + ib$ and k , where a , b and k are real and non-zero and $k < 0$. It is known that the graph of $y = Q(x)$ has two turning points.

- i. By a consideration of $Q'(x)$, show that $p < 0$. **1**

- ii. Deduce that $a > 0$. **2**

- iii. Show that $q = 8a^3 + 2ap$ **3**

c.



The base of the solid **K** shown in the diagram is the region in the x - y plane enclosed between the semi-ellipse $4x^2 + 9y^2 = 36$ and the y -axis. Each cross section perpendicular to the x -axis is an equilateral triangle.

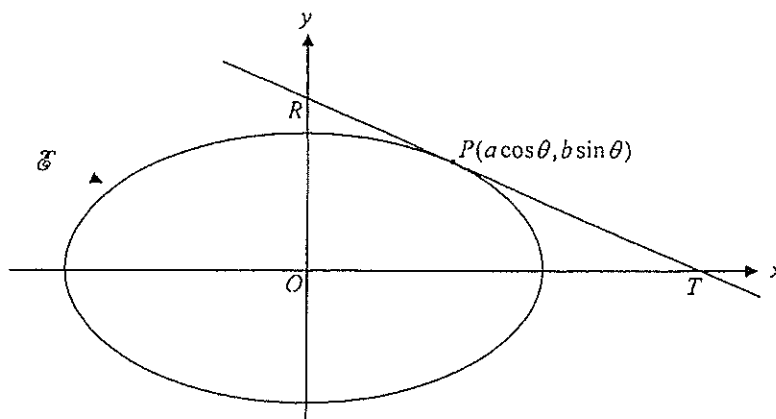
- i. Consider a slice of the solid with thickness Δx and distance x from the y -axis. Find the area of this slice in terms of x . **2**
- ii. Find the volume of the solid **K**. **2**
- iii. Solid **J** has the same base as solid **K** but its perpendicular cross sectional slice is an isosceles right angled triangle with its hypotenuse in the x - y plane. Find the ratio of volumes of solid **K** to solid **J**. **2**

Question 8 (15 marks) Commence each question on a SEPARATE page

- a. A particle P of mass m moves with constant angular velocity ω on a circle of radius r . Its position at time t is given by: $x = r \cos \theta$ $y = r \sin \theta$, where $\theta = \omega t$. **3**

Show that there is an inward radial force of magnitude $m r \omega^2$ acting on P .

b.



The ellipse E with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ shown in the diagram above, has a tangent at the point $P(a \cos \theta, b \sin \theta)$. The tangent cuts the x -axis at T and the y -axis at R .

- i. Show that the equation of the tangent at the point P is **2**

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1.$$

- ii. If T is the point of intersection between the tangent at point P and one of the directrices of the ellipse, show that $\cos \theta = e$. **2**

- iii. Hence find the angle that the focal chord through P makes with the x -axis. **1**

- iv. Using similar triangles or otherwise, show that $RP = e^2 RT$. **3**

- c. Let $I_n = \int_0^1 x(x^2 - 1)^n dx$ for $n = 0, 1, 2, \dots$

Use integration by parts to show that $I_n = \frac{-n}{n+1} I_{n-1}$ for $n \geq 1$. **4**

Standard integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2}).$$

$$\text{Note: } \ln x = \log_e x, \quad x > 0$$

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Question 1

$$a. \int \frac{dx}{x \ln x}$$

$$\text{Let } u = \ln x$$

$$\frac{du}{dx} = \frac{1}{x}$$

$$\therefore dx = x du$$

$$= \int \frac{x du}{x u}$$

$$= \int \frac{1}{u} du$$

$$= \ln(\ln x) + c$$

2

$$b. \int \frac{dx}{\sqrt{3+2x-x^2}}$$

$$= \int \frac{dx}{\sqrt{4-1+2x-x^2}}$$

$$= \int \frac{dx}{\sqrt{4-(x^2-2x+1)}}$$

$$= \int \frac{dx}{\sqrt{4-(x-1)^2}}$$

$$= \sin^{-1} \frac{x-1}{2} + c$$

2

$$c. \int \frac{dx}{(x+1)(x^2+4)}$$

$$\text{Now, } \frac{1}{(x+1)(x^2+4)} = \frac{a}{x+1} + \frac{bx+c}{x^2+4}$$

$$\therefore a(x^2+4) + (bx+c)(x+1) = 1$$

$$\text{Let } x = -1 \therefore 5a + 0 = 1$$

$$a = \frac{1}{5}$$

$$x = 0 \therefore 4a + c(1) = 1$$

$$\frac{4}{5} + c = 1$$

$$c = \frac{1}{5}$$

$$x = 1 \therefore 5\left(\frac{1}{5}\right) + (b + \frac{1}{5})(2) = 1$$

$$1 + 2b + \frac{2}{5} = 1$$

$$2b = -\frac{2}{5}$$

$$b = -\frac{1}{5}$$

$$\therefore \int \frac{\frac{1}{5}}{x+1} + \frac{-\frac{1}{5}x + \frac{1}{5}}{x^2+4} dx$$

$$= \frac{1}{5} \int \frac{1}{x+1} + \frac{1-x}{x^2+4} dx$$

$$= \frac{1}{5} \int \frac{1}{x+1} + \frac{1}{x^2+4} - \frac{x}{x^2+4} dx$$

$$= \frac{1}{5} \left[\ln|x+1| + \frac{1}{2} \tan^{-1} \frac{x}{2} - \frac{1}{2} \ln|x^2+4| \right] + c$$

$$d. \int \frac{15}{17+8\cos x} dx$$

$$\text{As } t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$dt = \frac{1}{2} (1+t^2) dx$$

$$\therefore dx = \frac{2 dt}{1+t^2}$$

$$\text{Also, } \cos x = \frac{1-t^2}{1+t^2}$$

$$\therefore \int \frac{15}{17+8\left(\frac{1-t^2}{1+t^2}\right)} \cdot \frac{2 dt}{1+t^2}$$

$$= \int \frac{30 dt}{17+17t^2+8-8t^2}$$

$$= \int \frac{30 dt}{25+9t^2}$$

$$= \frac{30}{9} \int \frac{dt}{\frac{25}{9} + t^2}$$

$$= \frac{30}{9} \cdot \frac{3}{5} \tan^{-1} \frac{3t}{5} + c$$

$$= 2 \tan^{-1} \frac{3t}{5} + c$$

3

$$e. i. \frac{d}{dx} \left[\frac{x}{\sqrt{x-3}} \right]$$

$$= \frac{\sqrt{x-3} \cdot 1 - \frac{1}{2}(x-3)^{-\frac{1}{2}} \cdot x}{x-3}$$

$$= \frac{\sqrt{x-3} - \frac{x}{2\sqrt{x-3}}}{x-3}$$

$$= \frac{x-3 - \frac{x}{2}}{(x-3)\sqrt{x-3}}$$

$$= \frac{2x-6-x}{2(x-3)\sqrt{x-3}}$$

$$= \frac{x-6}{2(x-3)\sqrt{x-3}}$$

$$\text{ii. } \int_4^7 \frac{2x-9}{2(x-3)\sqrt{x-3}} dx$$

$$= \int_4^7 \frac{x-6}{2(x-3)\sqrt{x-3}} dx + \int_4^7 \frac{x-3}{2(x-3)\sqrt{x-3}} dx$$

$$= \left[\frac{x}{\sqrt{x-3}} \right]_4^7 + \frac{1}{2} \int_4^7 (x-3)^{-\frac{1}{2}} dx$$

$$= \left(\frac{7}{2} - 4 \right) + \frac{1}{2} \left[\frac{(x-3)^{\frac{1}{2}}}{\frac{1}{2}} \right]_4^7$$

$$= -\frac{1}{2} + (2-1)$$

$$= -\frac{1}{2} + 1$$

$$= \frac{1}{2}$$

3

Question 2

$$\text{a. i. } r = \sqrt{(-1)^2 + (\sqrt{3})^2}$$

$$= 2$$

$$\tan \theta = \frac{\sqrt{3}}{-1}$$

$$\therefore \theta = \frac{2\pi}{3}$$

$$\therefore -1 + \sqrt{3}i = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)^2$$

$$\text{ii. } (-1 + \sqrt{3}i)^{-6}$$

$$= 2^{-6} \left[\cos -6 \left(\frac{2\pi}{3} \right) + i \sin -6 \left(\frac{2\pi}{3} \right) \right]$$

$$= \frac{1}{64} \left[\cos (-4\pi) + i \sin (-4\pi) \right]$$

$$= \frac{1}{64} [1 + i \cdot 0]$$

$$= \frac{1}{64}$$

2

$$\text{b. Let } z = x + iy$$

$$\therefore z + \frac{1}{z} = x + iy + \frac{1}{x+iy} \times \frac{x-iy}{x-iy}$$

$$= x + iy + \frac{x-iy}{x^2+y^2}$$

$$= \left[x + \frac{x}{x^2+y^2} \right] + i \left[y - \frac{y}{x^2+y^2} \right]$$

Now if $z + \frac{1}{z}$ is real, then

$$y - \frac{y}{x^2+y^2} = 0$$

$$\therefore y(x^2+y^2) - y = 0$$

$$\therefore y(x^2+y^2-1) = 0$$

$$\text{ie } y = 0 \quad \text{or} \quad x^2+y^2 = 1$$

$$\text{but } \text{Im}(z) = y$$

$$\therefore \text{Im}(z) = 0$$

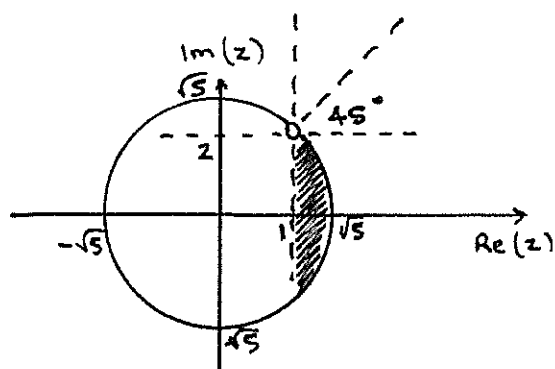
$$\text{but } |z| = \sqrt{x^2+y^2}$$

$$\therefore |z| = \sqrt{1}$$

$$\therefore |z| = 1$$

3

c.



3

$$\text{d. Let } z = \cos \theta + i \sin \theta$$

$$\therefore |1-z| = |1 - \cos \theta - i \sin \theta|$$

$$= \sqrt{(1 - \cos \theta)^2 + (-\sin \theta)^2}$$

$$= \sqrt{1 - 2\cos \theta + \cos^2 \theta + \sin^2 \theta}$$

$$= \sqrt{2 - 2\cos \theta}$$

$$|1+z| = |1 + \cos \theta + i \sin \theta|$$

$$= \sqrt{(1 + \cos \theta)^2 + \sin^2 \theta}$$

$$= \sqrt{1 + 2\cos \theta + \cos^2 \theta + \sin^2 \theta}$$

$$= \sqrt{2 + 2\cos \theta}$$

3

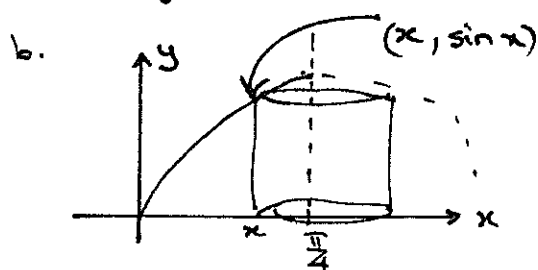
$$\therefore |1-z| = \sqrt{2-2\cos\theta} \quad |1+z| = \sqrt{2+2\cos\theta}$$

$$\begin{aligned} \text{ii. } \left| \frac{2}{1-z^2} \right| &= \frac{2}{|1-z||1+z|} \\ &= \frac{2}{\sqrt{2-2\cos\theta} \cdot \sqrt{2+2\cos\theta}} \\ &= \frac{2}{\sqrt{(2-2\cos\theta)(2+2\cos\theta)}} \\ &= \frac{2}{2\sqrt{1-\cos^2\theta}} \\ &= \frac{1}{\sin\theta} \\ &= \operatorname{cosec}\theta \end{aligned}$$

Question 3

$$\text{a. } \int_0^{\pi/4} x \sin x \, dx \quad \begin{array}{l} u=x \\ u'=1 \\ v'=\sin x \\ v=-\cos x \end{array}$$

$$\begin{aligned} \therefore \int uv' \, dx &= uv - \int u'v \, dx \\ &= -x \cos x \Big|_0^{\pi/4} + \int_0^{\pi/4} \cos x \, dx \\ &= -\frac{\pi}{4} \cdot \cos \frac{\pi}{4} - 0 + \sin x \Big|_0^{\pi/4} \\ &= -\frac{\pi}{4} \cdot \frac{1}{\sqrt{2}} + [\sin \frac{\pi}{4} - \sin 0] \\ &= -\frac{\pi}{4\sqrt{2}} + \frac{1}{\sqrt{2}} \\ &= \left(-\frac{\pi}{4} + 1 \right) \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{\sqrt{2}}{2} \left(1 - \frac{\pi}{4} \right) \\ &= \frac{\sqrt{2}}{8} (4 - \pi) \end{aligned}$$



$\therefore$ Volume of shell is δV , with thickness δx , height $y = \sin x$

$$\therefore \text{circumference} = 2\pi \left(\frac{\pi}{4} - x \right) \sin x \, \delta x$$

$$\therefore \delta V = 2\pi \left(\frac{\pi}{4} - x \right) \sin x \, \delta x$$

$$\begin{aligned} \therefore V &= 2\pi \int_0^{\pi/4} \left(\frac{\pi}{4} - x \right) \sin x \, dx \\ &= 2\pi \left[\frac{\pi}{4} \int_0^{\pi/4} \sin x \, dx - \int_0^{\pi/4} x \sin x \, dx \right] \\ &= \frac{\pi^2}{2} \int_0^{\pi/4} \sin x \, dx - 2\pi \int_0^{\pi/4} x \sin x \, dx \\ &= \frac{\pi^2}{2} [-\cos x]_0^{\pi/4} - 2\pi \cdot \frac{\sqrt{2}}{8} (4 - \pi) \\ &= \frac{\pi^2}{2} \left[-\frac{1}{\sqrt{2}} + 1 \right] - \frac{\sqrt{2}\pi}{4} (4 - \pi) \\ &= \frac{-\pi^2}{2\sqrt{2}} + \frac{\pi^2}{2} - \sqrt{2}\pi + \frac{\sqrt{2}\pi^2}{4} \\ &= -\frac{\sqrt{2}\pi^2}{4} + \frac{\pi^2}{2} - \sqrt{2}\pi + \frac{\sqrt{2}\pi^2}{4} \\ &= \frac{\pi}{2} (\pi - 2\sqrt{2}) \text{ units}^3 \end{aligned}$$

$$\text{c. } 9x^2 - 4y^2 = 36$$

$$\therefore \frac{x^2}{4} - \frac{y^2}{9} = 1$$

$$\therefore a=2 \text{ and } b=3$$

$$\therefore b^2 = a^2 (e^2 - 1)$$

$$\therefore 9 = 4(e^2 - 1)$$

$$e^2 - 1 = \frac{9}{4}$$

$$e^2 = \frac{13}{4}$$

$$e = \frac{\sqrt{13}}{2}$$

$$\therefore \text{foci } (\sqrt{13}, 0) \text{ and } (-\sqrt{13}, 0)$$

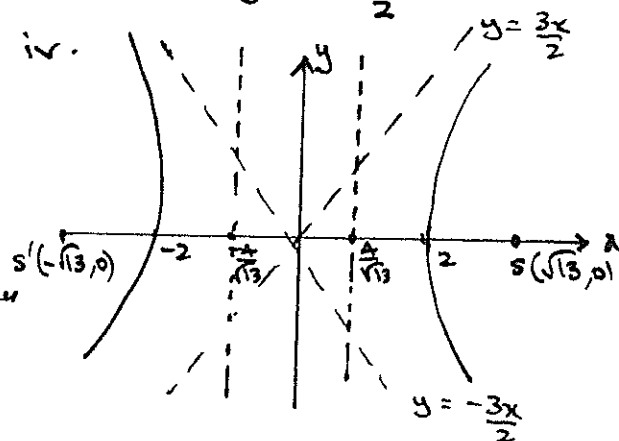
ii. Directrices:

$$x = \frac{4}{\sqrt{13}}, \quad x = -\frac{4}{\sqrt{13}}$$

iii. Asymptotes:

$$y = \pm \frac{3x}{2}$$

iv.



v. $9x^2 - 4y^2 = 36$

Differentiate with respect to x :

$$18x - 8y \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{18x}{8y}$$

At $P(x_0, y_0)$: $m = \frac{18x_0}{8y_0} = \frac{9x_0}{4y_0}$

$$\therefore y - y_0 = \frac{9x_0}{4y_0} (x - x_0)$$

$$4y_0 y - 4y_0^2 = 9x_0 x - 9x_0^2$$

$$\therefore 9x_0 x - 4y_0 y = 9x_0^2 - 4y_0^2 \quad \text{--- (1)}$$

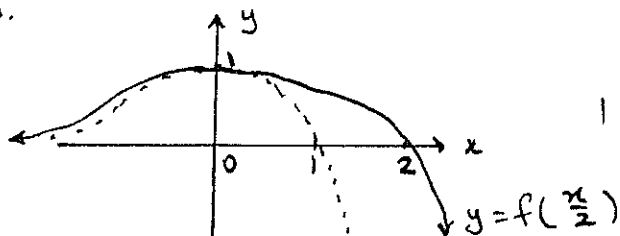
But as (x_0, y_0) lies on $9x^2 - 4y^2 = 36$

$$\therefore 9x_0^2 - 4y_0^2 = 36$$

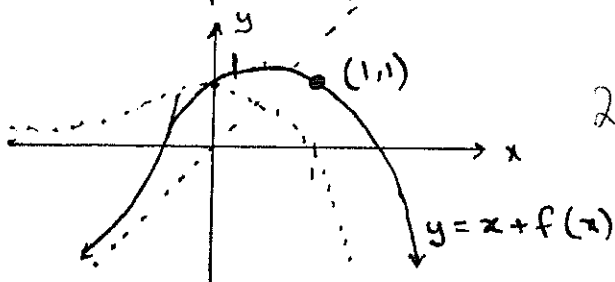
$$\therefore \textcircled{1} \text{ becomes } 9x_0 x - 4y_0 y = 36$$

Question 4:

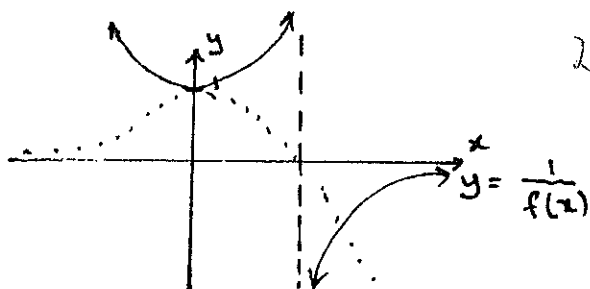
a. i.



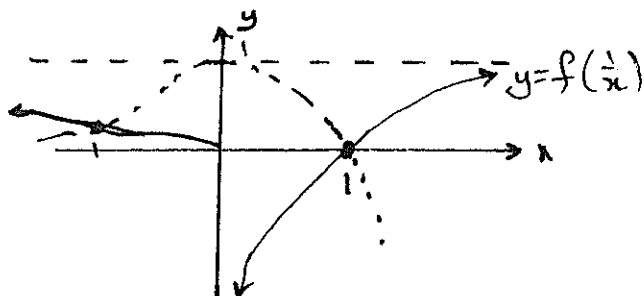
ii.



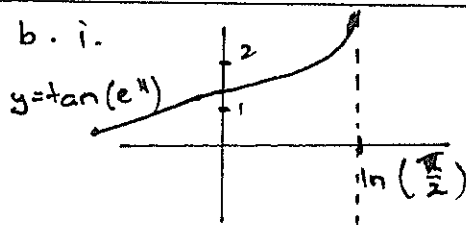
iii.



iv.



b. i.



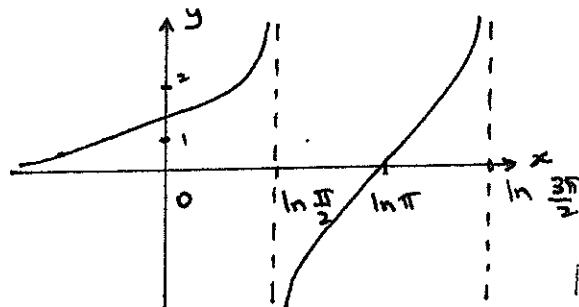
$$\tan e^x = 0$$

$$\therefore e^x = \pi, 2\pi, \dots$$

$$\therefore e^x = \pi$$

$$\therefore x = \ln \pi$$

ii.



iii. $\tan e^x = 0$

$$\therefore e^x = \pi, 2\pi, 3\pi, \dots$$

$$x = \ln \pi, \ln 2\pi, \dots$$

Now $\ln 6\pi = 2.93$ and $\ln 7\pi > 3$

$\therefore$ there are 6 solutions in the domain $1 < x < 3$.

iv. a. $y = \tan e^x$

$$\therefore x = \tan e^y$$

$$\therefore e^y = \tan^{-1} x$$

$$\therefore y = \ln(\tan^{-1} x)$$

β . When $\ln \frac{\pi}{2} < x < \ln \frac{3\pi}{2}$ then the equation of the original function is

$$y = \tan(e^x - \pi)$$

$$\therefore x = \tan(e^y - \pi)$$

$$e^y - \pi = \tan^{-1} x$$

$$e^y = \pi + \tan^{-1} x$$

$$y = \ln(\pi + \tan^{-1} x)$$

Question 5

a. i. Let $\alpha = \tan^{-1} n$
 $\tan \alpha = n$

and $\beta = \tan^{-1}(n-1)$

$\tan \beta = n-1$

$\therefore \tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$

$= \frac{n - (n-1)}{1 + n(n-1)}$

$= \frac{1}{1 + n^2 - n}$

$= \frac{1}{n^2 - n + 1}$

~~$\tan^{-1} \tan(\alpha - \beta) = \tan^{-1} \frac{1}{n^2 - n + 1}$~~

$\therefore \alpha - \beta = \tan^{-1} \frac{1}{n^2 - n + 1}$ 2

$\therefore \tan^{-1} - \tan^{-1}(n-1) = \tan^{-1} \frac{1}{n^2 - n + 1}$

ii. Let $n=1$

$\therefore \tan^{-1} 1 - \tan^{-1} 0 = \tan^{-1} \frac{1}{1^2 - 1 + 1}$

$\therefore \tan^{-1} 1 = \tan^{-1} 1$

Let $n=2$

$\therefore \tan^{-1} 2 - \tan^{-1} 1 = \tan^{-1} \frac{1}{2^2 - 2 + 1}$
 $= \tan^{-1} \frac{1}{3}$

Let $n=3$

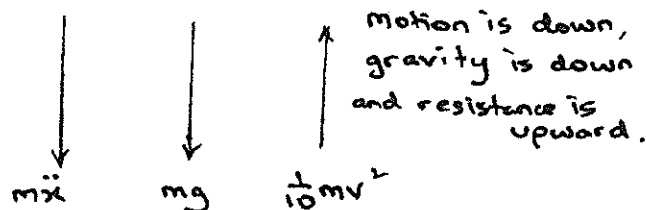
$\therefore \tan^{-1} 3 - \tan^{-1} 2 = \tan^{-1} \frac{1}{3^2 - 3 + 1}$
 $= \tan^{-1} \frac{1}{7}$

$\therefore \tan^{-1} 1 + \tan^{-1} \frac{1}{3} + \dots + \tan^{-1} \frac{1}{n^2 - n + 1}$ 2

$= \cancel{\tan^{-1} 1} + \cancel{\tan^{-1} 2} - \cancel{\tan^{-1} 1} + \cancel{\tan^{-1} 3}$
 $- \cancel{\tan^{-1} 2} + \dots + \cancel{\tan^{-1} n} + \cancel{\tan^{-1}(n-1)}$
 $= \tan^{-1} n$

iii. $\sum_{n=1}^{\infty} \tan^{-1} \frac{1}{n^2 - n + 1} = \lim_{n \rightarrow \infty} \tan^{-1} n$
 $= \frac{\pi}{2}$ 1

b. i.



$\therefore m\ddot{x} = mg - \frac{1}{10}mv^2$

$\therefore \ddot{x} = g - \frac{v^2}{10}$

$\therefore \ddot{x} = 10 - \frac{v^2}{10}$

$\frac{dv}{dt} = 10 - \frac{v^2}{10}$

$= \frac{100 - v^2}{10}$

$\frac{dt}{dv} = \frac{10}{100 - v^2}$

$t = \int \frac{10}{100 - v^2} dv$

$= 10 \int \frac{1}{(10-v)(10+v)} dv$

Now, by partial fractions:

$\frac{a}{10-v} + \frac{b}{10+v}$

$= a(10+v) + b(10-v) = 1$

$v = -10 \therefore 20b = 1 \therefore b = \frac{1}{20}$

$v = 10 \therefore 20a = 1 \therefore a = \frac{1}{20}$

$\therefore t = 10 \int \frac{1}{20(10-v)} + \frac{1}{20(10+v)} dv$

$= \frac{1}{2} [-\log(10-v) + \log(10+v)] + c$

$= \frac{1}{2} \left[\log \frac{10+v}{10-v} \right] + c$

$t=0, v=0 \therefore 0 = \frac{1}{2} \log 1 + c$
 $c=0$

$\therefore t = \frac{1}{2} \left[\log \frac{10+v}{10-v} \right]$

$\frac{10+v}{10-v} = e^{2t}$

$10+v = e^{2t}(10-v)$

$v + ve^{2t} = 10e^{2t} - 10$ 2

$v = \frac{10(e^{2t} - 1)}{e^{2t} + 1}$

$$v \frac{dv}{dx} = 10 - \frac{v^2}{10}$$

$$\therefore \frac{dv}{dx} = \frac{10}{v} - \frac{v}{10}$$

$$= \frac{100 - v^2}{10v}$$

$$\frac{dx}{dv} = \frac{10v}{100 - v^2}$$

$$x = -5 \ln(100 - v^2) + c$$

$$x=0, v=0$$

$$\therefore 0 = -5 \ln 100 + c$$

$$c = 5 \ln 100$$

$$\therefore x = 5 [\ln 100 - \ln(100 - v^2)]$$

$$x = 5 \ln \frac{100}{100 - v^2} \quad 2$$

$$\text{ii. } t = \frac{1}{2} \ln 99$$

$$\therefore v = 10 \left[\frac{e^{2 \cdot \frac{1}{2} \ln 99} - 1}{e^{2 \cdot \frac{1}{2} \ln 99} + 1} \right]$$

$$= 10 \left[\frac{99 - 1}{99 + 1} \right]$$

$$= \frac{98}{10}$$

$$= 9.8 \quad \therefore 9.8 \text{ ms}^{-1}$$

$$x = 5 \ln \left[\frac{100}{100 - 9.8^2} \right]$$

$$= 5 \ln \frac{2500}{99} \text{ metres} \quad 2$$

iii. After parachute disintegrates, only gravity is acting.

$$\therefore \ddot{x} = 10$$

$$\therefore v \frac{dv}{dx} = 10$$

$$\frac{dv}{dx} = \frac{10}{v}$$

$$\frac{dx}{dv} = \frac{v}{10}$$

$$\therefore x = \frac{v^2}{20} + c$$

$$\text{From ii, } v = 9.8, x = 5 \ln \frac{2500}{99}$$

$$\therefore 5 \ln \frac{2500}{99} = \frac{9.8^2}{20} + c$$

$$\therefore c = 5 \ln \frac{2500}{99} - 4.802$$

$$\therefore x = \frac{v^2}{20} + 5 \ln \frac{2500}{99} - 4.802$$

$$\text{When } x=100$$

$$\therefore 100 = \frac{v^2}{20} + 5 \ln \frac{2500}{99} - 4.802$$

$$\therefore v = 42 \quad (2 \text{ sig figs})$$

$$\therefore 42 \text{ ms}^{-1} \quad 3$$

Question 6

a. As $P(z)$ has real coefficients and $z-2+i$ is factor, then $z-2-i$ is also factor (conjugate pairs)

$\therefore (z-2+i)(z-2-i)$ is also factor

Now, as $2-i$ and $2+i$ are roots $\therefore$ sum of roots: 4

product of roots: 5

$\therefore z^2 - 4z + 5$ is a factor

$$\begin{array}{r} \overline{z^2 + 2z + 2} \\ \therefore z^2 - 4z + 5 \overline{2z^4 - 2z^3 - z^2 + 2z + 10} \\ \overline{2z^4 - 4z^3 + 5z^2} \\ \overline{2z^3 - 6z^2 + 2z} \\ \overline{2z^3 - 8z^2 + 10z} \\ \overline{2z^2 - 8z + 10} \\ \overline{0} \end{array}$$

$$\therefore P(z) = (z^2 - 4z + 5)(z^2 + 2z + 2) \quad 3$$

$$\text{b. } \ddot{x} = -9x + \frac{5}{(2-x)^2}$$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -9x + \frac{5}{(2-x)^2}$$

$$\frac{1}{2} v^2 = \frac{-9x^2}{2} + 5 \frac{(2-x)^{-1}}{-1 \cdot -1} + c$$

$$\frac{1}{2} v^2 = \frac{-9x^2}{2} + \frac{5}{2-x} + c$$

$$v=0, x=0$$

$$\therefore 0 = 0 + \frac{5}{2} + c \quad \therefore c = -\frac{5}{2}$$

$$\frac{1}{2} v^2 = \frac{-9x^2}{2} + \frac{5}{2-x} - \frac{5}{2}$$

$$v^2 = -9x^2 + \frac{10}{2-x} - 5$$

$$\therefore v^2 = \frac{-9x^2(2-x) + 10 - 5(2-x)}{2-x}$$

$$= \frac{-18x^2 + 9x^3 + 10 - 10 + 5x}{2-x}$$

$$= \frac{9x^3 - 18x^2 + 5x}{2-x}$$

$$= \frac{x(9x^2 - 18x + 5)}{2-x}$$

$$= \frac{x(3x-1)(3x-5)}{2-x} \quad 2$$

ii. Now, $v^2 = 0 \therefore x = 0, \frac{1}{3}, \frac{5}{3}$ and v^2 is undefined when $x = 2$.

Now, motion is possible only when $v^2 \geq 0 \therefore x(3x-1)(3x-5) \geq 0$



$$\therefore 0 \leq x \leq \frac{1}{3} \text{ and } \frac{5}{3} \leq x < 2$$

But, initially $x = 0$ 4

$\therefore$ travel between $x = 0$ and $x = \frac{1}{3}$

c. i. $\dot{x} = V + 3V \cos \theta$

$$\therefore x = Vt + 3Vt \cos \theta + c_1$$

$$x = 0, t = 0 \therefore 0 = 0 + 0 + c_1 \therefore c_1 = 0$$

$$\therefore x = Vt + 3Vt \cos \theta$$

Also, $\dot{y} = 3V \sin \theta - gt$

$$y = 3Vt \sin \theta - \frac{1}{2}gt^2 + c_2$$

$$y = 0, t = 0 \therefore 0 = 0 - 0 + c_2 \therefore c_2 = 0$$

$$\therefore y = 3Vt \sin \theta - \frac{1}{2}gt^2 \quad 2$$

ii. Range $\therefore y = 0$

$$\therefore 3Vt \sin \theta - \frac{1}{2}gt^2 = 0$$

$$t(3V \sin \theta - \frac{1}{2}gt) = 0$$

$$t = 0, t = \frac{3V \sin \theta}{\frac{1}{2}g}$$

$$= \frac{6V \sin \theta}{g}$$

$\therefore$ Subst in x

$$\text{i.e. } x = Vt + 3Vt \cos \theta$$

$$= t(V + 3V \cos \theta)$$

$$= \frac{6V \sin \theta}{g} \cdot V(1 + 3 \cos \theta)$$

$$= \frac{6V^2 \sin \theta}{g} (1 + 3 \cos \theta) \quad 2$$

$$\text{iii. } \frac{dR}{d\theta} = \frac{6V^2 \sin \theta}{g} (-3 \sin \theta) + (1 + 3 \cos \theta) \cdot \frac{6V^2 \cos \theta}{g}$$

$$= \frac{6V^2}{g} [-3 \sin^2 \theta + \cos \theta + 3 \cos^2 \theta]$$

$$= \frac{6V^2}{g} [-3(1 - \cos^2 \theta) + \cos \theta + 3 \cos^2 \theta]$$

$$= \frac{6V^2}{g} [-3 + 3 \cos^2 \theta + \cos \theta + 3 \cos^2 \theta]$$

$$= \frac{6V^2}{g} [6 \cos^2 \theta + \cos \theta - 3]$$

$$\text{Now } \frac{dR}{d\theta} = 0 \therefore 6 \cos^2 \theta + \cos \theta - 3 = 0$$

$$\therefore \cos \theta = \frac{-1 \pm \sqrt{1 - 4(6)(-3)}}{12} = \frac{-1 \pm \sqrt{73}}{12} \quad \text{But } \cos \theta > 0$$

$$\therefore \theta = \cos^{-1} \left[\frac{\sqrt{73} - 1}{12} \right]$$

Now check max/min

θ	$\cos^{-1} \frac{\sqrt{73}-1}{12}$	$\cos^{-1} \frac{\sqrt{73}-1}{12}$	$\cos^{-1} \frac{\sqrt{73}-1}{12}$
$\frac{dR}{d\theta}$	+	0	-

$\therefore \curvearrowright$

$$\therefore \text{Max when } \theta = \cos^{-1} \left[\frac{\sqrt{73}-1}{12} \right]$$

Question 7:

a. $p(x) = x^4 - 5x^3 - 9x^2 + 81x - 108$

Let triple root be α , other root is β

$$\therefore p'(x) = 4x^3 - 15x^2 - 18x + 81$$

$$p''(x) = 12x^2 - 30x - 18 = 0$$

$$2x^2 - 5x - 3 = 0$$

$$(2x+1)(x-3) = 0$$

$$x = -\frac{1}{2}, 3$$

$$\therefore \alpha = -\frac{1}{2} \text{ or } 3$$

Now, $p(-\frac{1}{2}) \neq 0 \therefore \alpha = 3$

Now, sum of roots:

$$3(\alpha) + \beta = 5$$

$$9 + \beta = 5$$

$$\beta = -4$$

$\therefore$ roots are 3, 3, 3, -4 3

b. i. $Q(x) = x^3 + px + q$, $Q'(x) = 3x^2 + p$

Now, as $Q(x)$ has 2 turning points,

$\therefore Q'(x)$ has 2 distinct real solns.

$$\therefore 3x^2 + p = 0$$

$$3x^2 = -p$$

$$x^2 = -\frac{p}{3}$$

$\therefore p$ must be negative,
ie $p < 0$

ii. As p, q real $\therefore a - ib$ is also root

$\therefore$ sum of roots: $a + ib + a - ib + k = 0$

$$2a + k = 0$$

$$a = -\frac{k}{2} \quad \text{--- ①}$$

But $k < 0 \therefore a > 0$ 2

iii. Product of roots = $-q$

$$\therefore k(a + ib)(a - ib) = -q$$

$$\therefore k(a^2 + b^2) = -q \quad \text{--- ②}$$

Also, roots 2 at a time:

$$\therefore k(a + ib) + k(a - ib) + (a + ib)(a - ib) = p$$

$$ka + ibk + ka - ibk + a^2 + b^2 = p$$

$$2ka + a^2 + b^2 = p$$

$$\therefore b^2 = p - 2ka - a^2 \quad \text{--- ③}$$

$$\text{From ① } k = -2a \quad \text{--- ④}$$

Subs ④ and ③ in ②

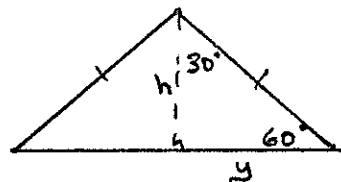
$$\therefore -2a(a^2 + p - 2a(-2a) - a^2) = -q$$

$$-2a(a^2 + p + 4a^2 - a^2) = -q$$

$$\therefore q = 2a(4a^2 + p)$$

$$\therefore q = 8a^3 + 2ap$$
 3

c.
i.



Consider the triangle above, taken as a slice of the solid with thickness Δx

$$\therefore \tan 30 = \frac{y}{h} \quad \therefore h = \frac{y}{\tan 30}$$

$$\therefore h = \sqrt{3}y$$

$$\therefore \text{Area of slice} : \frac{1}{2} \times 2y \times \sqrt{3}y = y^2\sqrt{3}$$

$$\text{Now, } 4x^2 + 9y^2 = 36$$

$$\therefore y^2 = \frac{36 - 4x^2}{9}$$

$$\therefore \text{Area} = \sqrt{3} \left[\frac{36 - 4x^2}{9} \right] \text{ units}^2 \quad 2$$

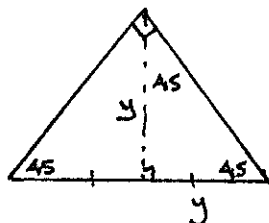
$$\text{ii. Vol} = \frac{\sqrt{3}}{9} \int_0^3 (36 - 4x^2) dx$$

$$= \frac{\sqrt{3}}{9} \left[36x - \frac{4x^3}{3} \right]_0^3$$

$$= \frac{\sqrt{3}}{9} [108 - 36 - 0]$$

$$= 8\sqrt{3} \text{ units}^3 \quad 2$$

iii.



$\therefore$ Area of slice

$$= \frac{1}{2} \times 2y \times y$$

$$= y^2$$

$$\therefore \text{Vol} = \int_0^3 y^2 dx$$

$$= \int_0^3 \frac{36 - 4x^2}{9} dx$$

$$= \frac{1}{9} \int_0^3 (36 - 4x^2) dx$$

$$= \frac{1}{9} \left[36x - \frac{4x^3}{3} \right]_0^3$$

$$= \frac{1}{9} [108 - 36]$$

$$= 8$$

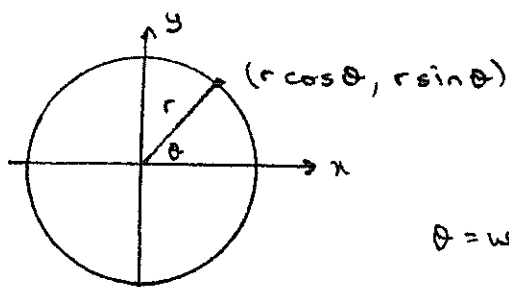
$\therefore$ Ratio of volume of K to volume of J

$$= 8\sqrt{3} : 8$$

$$= \sqrt{3} : 1$$
 2

Question 8:

a.



$$\theta = \omega t$$

$$x = r \cos \theta$$

$$= r \cos \omega t$$

$$y = r \sin \theta$$

$$= r \sin \omega t$$

$$\therefore \dot{x} = -r\omega \sin \omega t$$

$$\therefore \dot{y} = r\omega \cos \omega t$$

$$\ddot{x} = -r\omega^2 \cos \omega t$$

$$\ddot{y} = -r\omega^2 \sin \omega t$$

$$= -r\omega^2 \cos \theta$$

$$= -r\omega^2 \sin \theta$$

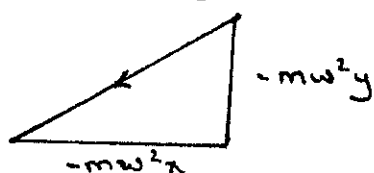
$$= -\omega^2 x$$

$$= -\omega^2 y$$

$$\therefore \text{Horizontal: } F_x = m \cdot -\omega^2 x$$

$$= -m\omega^2 x$$

$$\text{Vertical: } F_y = -m\omega^2 y$$



$$\therefore F = \sqrt{(m\omega^2 x)^2 + (m\omega^2 y)^2}$$

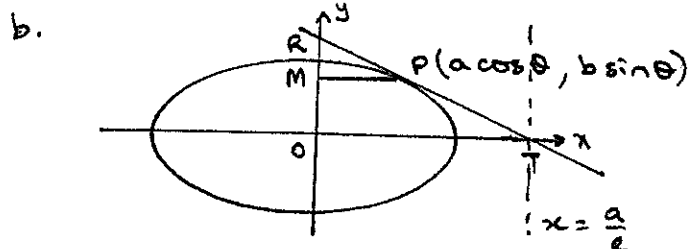
$$= \sqrt{m^2 \omega^4 x^2 + m^2 \omega^4 y^2}$$

$$= m\omega^2 \sqrt{x^2 + y^2}$$

$$\therefore F = m\omega^2 r$$

3

Force vector directed inwards



i. Diff wrt x:

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{a^2} \times \frac{b^2}{xy}$$

$$= -\frac{b^2 x}{a^2 y}$$

At $(a \cos \theta, b \sin \theta)$,

$$m = \frac{-b^2 x \cos \theta}{a^2 y \sin \theta}$$

$$= \frac{-b \cos \theta}{a \sin \theta}$$

$$\therefore y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a y \sin \theta - a b \sin^2 \theta = -b x \cos \theta + a b \cos^2 \theta$$

$$b x \cos \theta + a y \sin \theta = a b (\sin^2 \theta + \cos^2 \theta)$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \quad \text{--- (1) 2}$$

ii. As T lies on an asymptote

$$\therefore T\left(\frac{a}{e}, 0\right)$$

$\therefore$ sub in (1)

$$\therefore \frac{\frac{a}{e} \cos \theta}{a} + 0 = 1$$

$$\frac{1}{e} \cos \theta = 1$$

$$\cos \theta = e$$

$$\cos \theta = e$$

2

iii. Focus $(ae, 0)$

and as $\cos \theta = e$

$$\therefore P(ae, b \sin \theta)$$

$\therefore$ focal chord is vertical

$\therefore$ makes 90° with x axis

iv. For R, let $x=0$ in tangent

$$\therefore \frac{y \sin \theta}{b} = 1$$

$$\therefore y = \frac{b}{\sin \theta} \therefore R\left(0, \frac{b}{\sin \theta}\right)$$

Let M lie on y-axis, where $PM \perp RM$

$$\therefore M(0, b \sin \theta)$$

$\therefore \triangle ROT \parallel \triangle RMP$

$$\therefore \frac{RP}{RT} = \frac{RM}{RO}$$

$$= \frac{b}{\sin \theta} - b \sin \theta$$

$$\frac{b}{\sin \theta}$$

$$= \frac{b - b \sin^2 \theta}{\sin \theta} \times \frac{\sin \theta}{b}$$

$$= 1 - \sin^2 \theta$$

$$= \cos^2 \theta = e^2 \therefore RP = e^2 RT$$

3

b. $\int_0^1 x(x^2-1)^n dx$

i.

$$\int uv' dx = uv - \int u'v dx$$

$$u = (x^2-1)^n$$

$$u' = n(x^2-1)^{n-1} \cdot 2x$$

$$v' = x$$

$$v = \frac{x^2}{2}$$

$$\therefore \left[\frac{x^2}{2} (x^2-1)^n \right]_0^1 - \int_0^1 \frac{x^2}{2} \cdot n(x^2-1)^{n-1} \cdot 2x dx$$

$$= \frac{1}{2}(0) - 0 - n \int_0^1 x^3 (x^2-1)^{n-1} dx$$

$$= -n \int_0^1 \frac{x^3 (x^2-1)^n}{x^2-1} dx$$

$$= -n \int_0^1 \frac{x^3}{x^2-1} [x^2-1]^n dx$$

$$\therefore x^2-1 \overline{) \begin{array}{r} x^3 + 0x^2 + 0x + 0 \\ \underline{x^3} \\ -x \\ \underline{-x} \\ x \end{array}}$$

$$= -n \int_0^1 \left(x + \frac{x}{x^2-1} \right) (x^2-1)^n dx$$

$$= -n \int_0^1 x(x^2-1)^n + \frac{x}{x^2-1} (x^2-1)^n dx$$

$$= -n \int_0^1 x(x^2-1)^n dx - n \int_0^1 x(x^2-1)^{n-1} dx$$

$$\therefore I_n = -n I_n - n I_{n-1} \quad \text{if}$$

$$\therefore (1+n) I_n = -n I_{n-1}$$

$$\therefore I_n = \frac{-n}{n+1} I_{n-1} \text{ for } n \geq 1$$